Imperial College London

Function (函数) **Sequences** (数列)

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40016 Calculus — CO-BEng/MEng Computing Year 1

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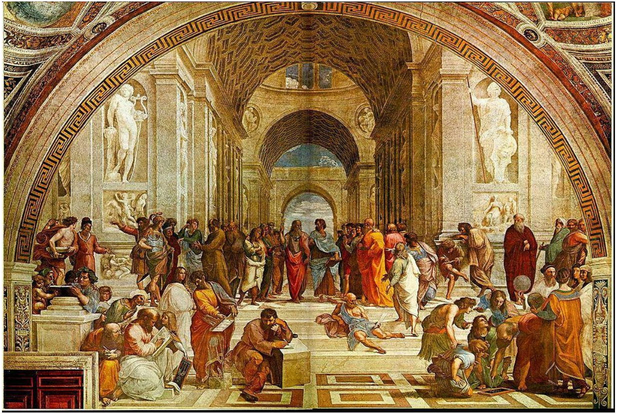
On questions



- The only stupid question is one not asked.

Binary relations

二元关系

我们可以自己定义这个关系

Set (集合): {a, b, c, ...}
a,b,c are the elements (元素)
- 集合里面的元素不能重复

(笛卡尔积)

- A binary relation over sets A and B is a subset of the Cartesian product $A \times B$; that is, it is a set of ordered pairs (a, b) consisting of elements a in A and b in B.
- It encodes the common concept of “relation”: an element a is related to an element b, if and only if the pair (a, b) belongs to the set of ordered pairs that defines the binary relation.
- A binary relation is the most studied special case $n = 2$ of an n-ary relation over sets $A_1, \ldots, A_n$, which is a subset of the Cartesian product $A_1 \times \ldots \times A_n$.

|A1 * A2 * .... * An|=|A1| * |A2| * .... * |An|

Example:
Set A: {3, 4, 5, ..., J, Q, K, A} |A|=13; |B|=4; |A * B| = 52
Set B: {D, B, ...}

(3, D)

Examples

Integer (整数，包括正数，负数，0) 质数 (2,3,5,7,...)

- An example of a binary relation is the "divides" relation over the set of prime numbers $\mathbb{P}$ and the set of integers $\mathbb{Z}$, in which each prime p is related to each integer z that is a multiple of p, but not to an integer that is not a multiple of p. multiple (倍数)
- In this relation, for instance, the prime number 2 is related to numbers such as $-4, 0, 6, 10$, but not to 1 or 9, just as the prime number 3 is related to 0, 6, and 9, but not to 4 or 13.
- We can put this in the language of sets: $(2, -4)$, $(2, 0)$, $(2, 6)$, and $(2, 10)$ are elements of this binary relation, whereas $(2, 1)$ and $(2, 9)$ are not; likewise, $(3, 0)$, $(3, 6)$, $(3, 9)$ are elements of this binary relation, but $(3, 4)$ and $(3, 13)$ are not.
- Binary relations are used in many branches of mathematics to model a wide variety of relationships. These include, among others:
  - the "is greater than," "is equal to," and "divides" relations in arithmetic;
  - (a+bi, a-bi) the "is congruent to" relation in geometry; if a > b, then we have (a,b); if a = b, then we have (a,b)
  - the "is adjacent to" relation in graph theory;
  - the "is orthogonal to" relation in linear algebra.

Orthogonal (正交) e.g., (a,b) if a * b = 0

Functions 函数

定义域 (domain field)

- A function f from a set A to a set B is a mathematical object that associates to every element a in A a unique element $f(a)$ in B.
- We denote this by $f : A \to B$.
- For example, $f : \mathbb{N} \to \mathbb{N}$ with $f : x \mapsto x + 1$ or $f(x) = x + 1$ is a function since it maps every natural number $x \geq 1$ to a unique natural number $x + 1$.
- In contrast, the mathematical object that associates to every element x in $\mathbb{N}$ both x and $-x$ is not a function $\mathbb{N} \to \mathbb{N}$ since it associates to 3, for example, two elements, 3 and -3.

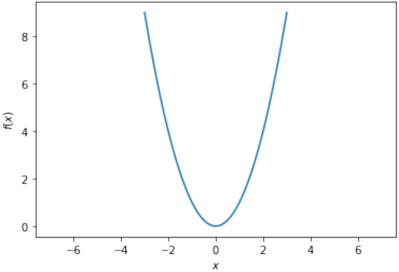
  3 -> { {empty}, {3}, {-3}, {3,-3}}
- ~~If we take the same map $x \mapsto \{-x, x\}$ and give it the type $\mathbb{N} \to 2^{\mathbb{Z}}$ where $2^{\mathbb{Z}}$ is the power set of $\mathbb{Z}$, the set of all subsets of $\mathbb{Z}$, then this is a function.~~
- A function is a binary relation and so functions are special binary relations.
- Not all binary relations are functions: for example "divides" associates to 2 both 4 and 6 and is therefore not a function; "equals" associates each element of a set (only) to itself and therefore is a function (called an identity function).

Power set: assume we have a set {1, 2, 3}, then the power set will be: (所有子集构成的集合)
{ {empty}, {1}, {2}, {3}, {1,2,}, {1,3,}, {2,3,}, {1,2,3} }

Another example

set A; Set B

- Here is another example of a function: $f : \mathbb{R} \to \mathbb{R}$, $f : x \mapsto x^2$.
- We can plot the graph of this function:



- This graph is called...

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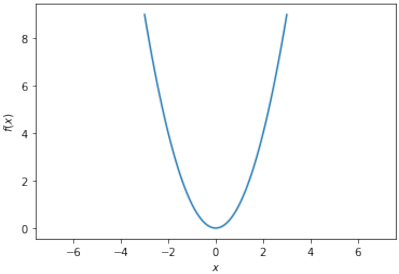


- This graph is called... a parabola.

Nota bene!

- Functions are a core concept in mathematics and computer science.

Functions in programming

- When we program, we can also define functions, for example

```
def square(x):
    return x * x
```

- We can then call (evaluate) this function

```
square(3)
```

  and obtain the answer 9.

- In programming, functions can take multiple values as arguments (parameters), for example

```
def add(x, y):
    return x + y
```

```
add(3, 5)
```

  (that gives the answer 8).

- This function corresponds to the mathematical function of type add : $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$.

Side effects

- In programming, functions can have side effects; for example, they can output text to the console (write a record to a database, send data over the network, etc.):

```
def greet(name):
    print('Hello,', name)

greet('Paul')
```

```
Hello, Paul
```

- In mathematics we don't consider functions with side effects. We are solely concerned with the input and output of a function; a mathematical function can have no side effects.

Sequences

定义域现在变为自然数

- We begin this course with the study of functions of the type $f : \mathbb{N} \to \mathbb{R}$ that associates real numbers to natural numbers.
- We will call these functions sequences, and the outputs $f(n)$ will be called terms of such sequences.

项

Formal definition

N: 自然数(包括o)；R: 实数

1. A sequence is a function $f : \mathbb{N} \to \mathbb{R}$ that maps natural numbers to real numbers. Often it is convenient to define a sequence as a function $f : \mathbb{N}^+ \to \mathbb{R}$ that maps positive natural numbers to real numbers, which is the terminology we adopt in this course.
2. We will mostly write $(a_n)_{n \geq 1}$ for such a sequence, where $a_n = f(n)$. n is the index (项数)
3. By abuse of notation, we may write a_n or $a_1, a_2, \ldots$ for such a sequence if the context makes it clear that we refer to a sequence and not to one of its elements.

Examples

- $f : \mathbb{N}^+ \to \mathbb{R}$, $f : n \mapsto 1$, also written $(1)_{n \geq 1}$, is a sequence:

  converge; converge to 1

  $$1, 1, 1, 1, 1, \ldots.$$

- $f : \mathbb{N}^+ \to \mathbb{R}$, $f : n \mapsto \frac{1}{n}$, also written $(\frac{1}{n})_{n \geq 1}$, is a sequence:

  converge; converge to 0

  $$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots.$$

- $f : \mathbb{N}^+ \to \mathbb{R}$, $f : n \mapsto n^2 - n$, also written $(n^2 - n)_{n \geq 1}$, is also a sequence:

  converge; converge to infinite

  $$0, 2, 6, 12, 20, \ldots.$$

- $f : \mathbb{N}^+ \to \mathbb{R}$, $f : n \mapsto (-1)^n$, also written $((-1)^n)_{n \geq 1}$ is yet another sequence:

  diverge

  $$-1, 1, -1, 1, -1, \ldots.$$

- $f : \mathbb{N}^+ \to \mathbb{R}$ defined by

  diverge

  $$f(n) = \begin{cases} 10^{-6}, & \text{if } n \text{ is prime,} \\ \frac{1}{n}, & \text{if } n \text{ is not prime} \end{cases}$$

  is yet another sequence:

  $$1, 10^{-6}, 10^{-6}, \frac{1}{4}, 10^{-6}, \frac{1}{6}, 10^{-6}, \frac{1}{8}, \frac{1}{9}, \frac{1}{10}, 10^{-6}, \ldots.$$

Remark

- Note that $(a_n)_{n\geq 1}$ and $(a_i)_{i\geq 1}$ refer to the same sequence as index variables are bound parameters that do not change meaning.

Arithmetic sequence

Arithmetic (算术)； Arithmetic sequence (等差数列)

- An arithmetic sequence (also known as an arithmetic progression) is the sequence $f : \mathbb{N}^+ \to \mathbb{R}$ defined by

$$f : n \mapsto \begin{cases} a_1, & n = 1, \\ a_1 + (n-1)d, & \text{otherwise.} \end{cases}$$

  where a_1 is the first term and d the common difference.
- In general,

$$f(n) = f(m) + (n-m)d.$$

- Here is an example of an arithmetic sequence:

$$5, 7, 9, 11, 13, 15, \ldots.$$

- What are a_1 and d?
- $a_1 = 5$, $d = 2$.

The sum of the first n terms of an arithmetic sequence

- The sum of the first n terms of an arithmetic sequence can be written in several different ways:

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \ldots + (a_1 + (n-2)d) + (a_1 + (n-1)d),$$
$$S_n = (a_n - (n-1)d) + (a_n - (n-2)d) + \ldots + (a_n - 2d) + (a_n - d) + a_n.$$

- Adding both sides of the two equations, all terms involving d cancel:

$$2S_n = n(a_1 + a_n).$$

- Dividing both sides by 2 produces a common form of the equation:

$$S_n = \frac{n}{2}(a_1 + a_n).$$

The geometric sequence

Geometric sequence (等比数列)

- A geometric sequence (also known as a geometric progression) is the sequence $f : \mathbb{N}^+ \to \mathbb{R}$ defined by
$$f : n \mapsto ar^{n-1},$$
where a is a scale factor, equal to the sequence's first term, and r the common ratio.
- Thus the terms of this sequence are
$$a, ar, ar^2, ar^3, ar^4, \ldots.$$
- A geometric sequence follows the recursive relation
$$a_n = ra_{n-1}$$
for every integer $n \geq 2$.

Recursive / Recursion (递归)：一个函数在计算的时候使用到了自己

Fib n = (Fib n - 1) + (Fib n -2)

1,1,2,3,5,8,...

The sum of the first n terms of a geometric sequence

- Let us derive the sum of the first n terms of a geometric sequence,

$$S_n = ar^0 + ar^1 + ar^2 + \ldots + ar^{n-1}.$$

- Multiply both sides by $(1 - r)$:

$$\begin{aligned}(1-r)S_n &= (1-r)\left(ar^0 + ar^1 + ar^2 + \ldots + ar^{n-1}\right)\\ &= \left(ar^0 + ar^1 + ar^2 + \ldots + ar^{n-1}\right) - \left(ar^1 + ar^2 + ar^3 + \ldots + ar^n\right)\\ &= a - ar^n,\end{aligned}$$

since all the other terms cancel.

- If $r \neq 1$, we can rearrange the above to get the convenient formula

$$S_n = \frac{a(1-r^n)}{(1-r)}.$$

The Fibonacci sequence

- It is possible to specify a sequence recursively, where the latter terms are defined in terms of the previously defined terms.
- For example, the Fibonacci sequence $f : \mathbb{N}^+ \to \mathbb{R}$ is defined by

$$f : n \mapsto \begin{cases} 0, & n = 1, \\ 1, & n = 2, \\ f(n-1) + f(n-2), & n \geq 3. \end{cases}$$

- What are the first few terms of this sequence?
- $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$.
- We can easily find that this is sequence A000045 in the On-line Encyclopedia of Integer Sequences (OEIS): `https://oeis.org/A000045`

Monotonicity

Monotonicity (单调性) Non-monotonic / Partially Monotonic

(任意)

(递增) (递减)

- A sequence $(a_n)_{n\geq 1}$ is increasing if $a_{n+1} \geq a_n$ for $n \geq 1$; it is decreasing if $a_{n+1} \leq a_n$ for $n \geq 1$.
- The sequence is called monotonic if it is either increasing or decreasing. (不分段的)
- The notions of a strictly increasing ($a_{n+1} > a_n$ for $n \geq 1$) and strictly decreasing sequence ($a_{n+1} < a_n$ for $n \geq 1$) are also similarly defined.

Definition (定义)

Examples

Solution 1:
We have that for all n, a_n = 1/n, a_{n+1} = 1/(n +1), hence that
a_n - a_{n+1} = 1/n - 1/(n+1) = (n+1)-n / n*(n+1) = 1/n*(n+1) > 0 which is the same as a_n - a_{n+1} > 0
from the definition, the sequence is strictly decreasing

- The sequence $\left(\frac{1}{n}\right)_{n\geq 1}$ is strictly decreasing. It is therefore decreasing and monotonic.
- The sequence $(n^2 - n)_{n\geq 1}$ is strictly increasing. It is therefore increasing and monotonic.

Question

Is the Fibonacci sequence

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34$$

- increasing? yes
- decreasing? no
- strictly increasing? no
- strictly decreasing? no
- monotonic? yes

Question

- Why are sequences important?

Use case 1: approximate solution of equations (i)

- We learn how to solve some equations analytically at school. For example, we know how to solve a quadratic equation, such as $2x^2 + 4x - 4 = 0$, using the appropriate formula.
- However, how do we solve something like $\cos(x) = x^3$?
- Typically we use a numerical method, such as Newton's method, to solve such equations. 数值(计算)方法
- We will study Newton's method in detail later in this course. For now, let's just say that solving $\cos(x) = x^3$ is equivalent to finding the zero of $f(x) = \cos(x) - x^3$.
- We have $f'(x) = -\sin(x) - 3x^2$. (If this is your first encounter with a derivative, don't worry—in this course you will see many of them.)

Use case 1: approximate solution of equations (ii)

- Since $-1 \leq \cos(x) \leq 1$ for all x, whereas $x^3 < 1$ for $x < 1$ and $x^3 > 1$ for $x > 1$, we know that our solution must lie between -1 and 1.
- For example, with an initial guess $x_0 = 0.5$, the sequence (!) given by Newton's method is

$$
\begin{aligned}
x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} &&= 0.5 - \frac{\cos 0.5 - 0.5^3}{-\sin 0.5 - 3 \times 0.5^2} &&= 1.112\,141\,637\,097\ldots \\
x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} &&= \vdots &&= \underline{0.}909\,672\,693\,736\ldots \\
x_3 &= \vdots &&= \vdots &&= \underline{0.867}\,263\,818\,209\ldots \\
x_4 &= \vdots &&= \vdots &&= \underline{0.865\,477}\,135\,298\ldots \\
x_5 &= \vdots &&= \vdots &&= \underline{0.865\,474\,033\,1}11\ldots \\
x_6 &= \vdots &&= \vdots &&= \underline{0.865\,474\,033\,102}\ldots
\end{aligned}
$$

Numerical methods and sequences

- Thus a numerical method for finding roots (and minima, and maxima) gives us not the true solution, but a sequence of approximate solutions.
- This sequence (hopefully!) has a special property that the approximate solutions get closer and closer to the true solution.
- 收敛
  We nearly said "that the approximate solutions converge to the true solution", but we haven't defined that term yet.
- When we train a neural network (another numerical method), we find a sequence of approximate weights and biases, which hopefully get closer and closer to the globally optimal weights and biases (although often we end up getting stuck in a local optimum, which is unfortunate).

Use case 2: real numbers

- Every real number has a decimal expression. The decimal expression

$$b_0.b_1b_2b_3\ldots,$$

represents the real number that is the "sum to infinity" of the series

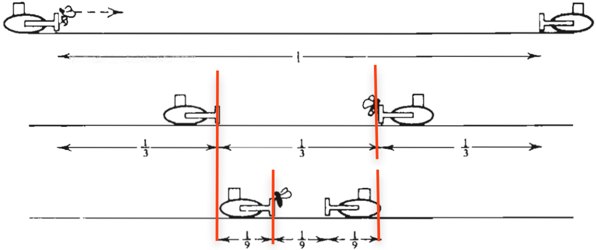
$$b_0 + \frac{b_1}{10} + \frac{b_2}{10^2} + \frac{b_3}{10^3} + \ldots.$$

- In other words, the real number $b_0.b_1b_2b_3\ldots$ is the sum of the sequence

$$b_0, \frac{b_1}{10}, \frac{b_2}{10^2}, \frac{b_3}{10^3}, \ldots.$$

- In order to make this statement precise, we need to introduce one of the most fundamental concepts in mathematics—that of the convergence to a limit of a sequence of real numbers.

Two bulldozers and the bee



- Two bulldozers are moving towards each other at a speed of one mile per hour on a collision course.
- When they started they were one mile apart and a bee was perched on the front of one of them.
- The bee began to fly back and forward between the bulldozers at a constant speed of two miles per hour, vainly seeking to avoid his fate.
- How far from his starting point will the unfortunate insect be crushed?

Quick answer

- It is quite obvious that the bee will be crushed when the bulldozers collide.
- Since the bulldozers travel at the same speed, this will happen halfway between their starting points.
- The answer is therefore one half mile.

Slow answer

- This riddle is usually posed in the hope that the victim will embark on some complicated calculations involving the flight of the bee.
- Let us take up the role of the riddler's victim and examine the flight of the bee.
- Let x_n denote the distance the bee is from his starting point when he makes his nth landing. Then

$$x_1 = \frac{2}{3},$$
$$x_2 = \frac{2}{3} - \frac{2}{9}.$$

- Question: what is x_n?
- Answer:

$$x_n = \frac{2}{3} - \frac{2}{9} + \frac{2}{27} - \ldots + (-1)^{n-1} 2 \left(\frac{1}{3}\right)^n.$$

- Can you simplify this expression?

On simplifying

- On simplifying,

$$
\begin{aligned}
x_n &= \frac{2}{3} - \frac{2}{9} + \frac{2}{27} - \ldots + (-1)^{n-1} 2 \left(\frac{1}{3}\right)^n \\
\text{等比数列} &= \frac{2}{3}\left\{1 + \left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right)^2 + \ldots + \left(-\frac{1}{3}\right)^{n-1}\right\} \\
&= \frac{2}{3}\left\{\frac{1-\left(-\frac{1}{3}\right)^n}{1-\left(-\frac{1}{3}\right)}\right\} = \frac{1}{2}\left(1 - \left(-\frac{1}{3}\right)^n\right).
\end{aligned}
$$

- How is the answer $\frac{1}{2}$ to be extracted from this formula?
- It seems clear that, as n gets larger and larger, $\left(-\frac{1}{3}\right)^n$ gets closer and closer to zero and so x_n gets closer and closer to $\frac{1}{2}$.

  收敛的基本思想(这个数列会逐渐的趋近于某一个数)

Take a deep breath...

- ...you are about to learn the most important definition in all of analysis.
- Understand it well.

Convergence to a limit, divergence

收敛的定义 (通过极限)

一个具体的收敛值

iff: 正着推，反着推都可以

Let $(a_n)_{n\geq 1}$ be a sequence.

1. The sequence $(a_n)_{n\geq 1}$ converges to a limit l in $\mathbb{R}$ iff the following statement can be shown to be true: 我们都可以在自然数里面找到一个N

   epsilon (误差)

   $$\text{For all } \epsilon > 0\text{, we can find a } N \text{ in } \mathbb{N} \text{ such that, for all } n > N: |a_n - l| < \epsilon. \qquad (1)$$

   In that case, we denote this fact as $\lim_{n\to\infty} a_n = l$ or $(a_n)_{n\geq 1} \to l$.
2. The sequence $(a_n)_{n\geq 1}$ converges to ∞ (in the extended real line $\mathbb{R} \cup \{\infty\}$) iff for all r in $\mathbb{R}$, there is an N in $\mathbb{N}$ such that for all $n \geq N$ we have $a_n > r$. We write $\lim_{n\to\infty} a_n = \infty$ or $(a_n)_{n\geq 1} \to \infty$ in that case.
3. The sequence $(a_n)_{n\geq 1}$ converges to $-\infty$ (in the extended real line $\mathbb{R} \cup \{-\infty\}$) iff for all r in $\mathbb{R}$, there is an N in $\mathbb{N}$ such that for all $n \geq N$ we have $a_n < r$. We write $\lim_{n\to\infty} a_n = -\infty$ or $(a_n)_{n\geq 1} \to -\infty$ in that case.
4. The sequence $(a_n)_{n\geq 1}$ converges if it converges either to a real number or to ∞ or to $-\infty$.
5. The sequence $(a_n)_{n\geq 1}$ diverges if it does not converge.

发散

Understanding the definition

- The statement about convergence is best understood by taking the last part first and working backwards.
- The limit inequality $|a_n - l| < \epsilon$ says that the distance from the nth term in the sequence to the limit should be less than ϵ.
- For all $n > N$... There should be some point N after which all the terms a_n in the sequence are within ϵ of the limit.
- For all $\epsilon > 0$... No matter what value of ϵ we pick, however tiny, we will still be able to find some value of N such that all the terms to the right of a_N in that sequence are within ϵ of the limit.

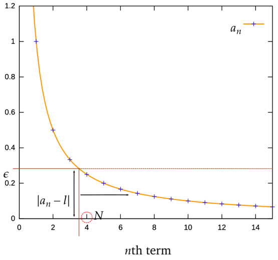
Examples

- The sequence $(n)_{n\geq 1}$ converges to ∞.
- The sequence $(-\ln n)_{n\geq 1}$ converges to $-\infty$.
- The sequence $((-1)^n \times n)_{n\geq 1}$ grows without bounds but it always jumps between negative and positive values in that growth. Therefore, this sequence diverges.

How do we prove convergence

- The above definition of convergence also tells us how to prove that a sequence converges to a real number l.
- Someone gives us a positive ϵ, we have no control over that value.
- For this value of ϵ, we have to then find a natural number N and demonstrate that $|a_n - l| < \epsilon$ for all $n > N$.
- It may be helpful to think of the person who gives you that ϵ acting in the role of a Refuter who claims that this sequence does not have limit l.
- So you, as the Verifier of the opposite claim (that the sequence has l as limit), then need to find a value N that will convince the Refuter.
- In particular, you cannot first determine a value N and then let the refuter choose ϵ.
- Rather, the value N is a function of the choice of ϵ.
- We can similarly formulate how we can prove the convergence of a sequence to ∞ or to $-\infty$.

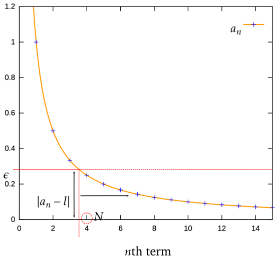
Illustration of convergence



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- Consider the sequence $a_n = \frac{1}{n}$ for $n \geq 1$.
- The game is to find a value of N for whatever value of ϵ is chosen.
- N represents the point after which all a_n will be within ϵ of the limit.
- Clearly the smaller the value of ϵ chosen, the further to the right we will have to go to achieve this, thus the larger the value of N we will have to choose.

Applying the limit inequality



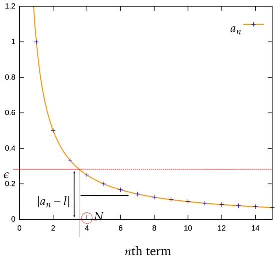
The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- We have taken a particular value of ϵ to demonstrate the concepts involved.
- Taking $\epsilon = 0.28$ (picked arbitrarily), we apply the $|a_n - l| < \epsilon$ inequality, to get

$$|1/n| < 0.28$$
$$1/n < 0.28$$
$$n > 1/0.28$$
$$n > 3.5714\ldots$$

- This states that for $n > 3.57$ the function $1/n$ will be less than 0.28 from the limit.

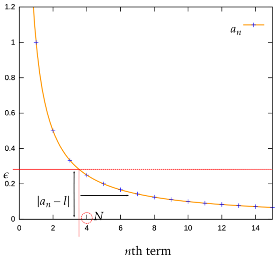
Which N to pick?



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- However, we are dealing with a sequence which is only defined at integer points on this function.
- Hence we need to find the Nth term, a_N, which is the first point in the sequence to also be less than this value of ϵ.
- In this case, we can see that the next term above 3.5714..., that is $N = 4$, satisfies this requirement.

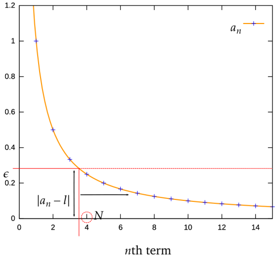
For all $n > N$



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- We are further required to ensure that all other points to the right of a_N in the sequence $(a_n)_{n>N}$ are also within 0.28 of the limit.
- We know this to be true because the condition on n we originally obtained was $n > 3.57$, so we can be sure that $a_N, a_{N+1}, a_{N+2}, \ldots$ are all closer to the limit than 0.28.

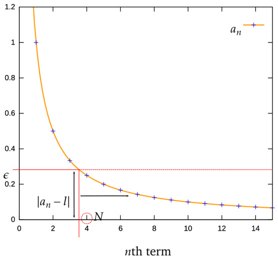
For all $\epsilon > 0$



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- The final point to note is that it is not sufficient to find N for just a single value of ϵ. We need to find N for every value of $\epsilon > 0$.
- Since N will vary with ϵ as mentioned above, we are clearly required to find a function of type $\mathbb{R}^+ \to \mathbb{N}$ that maps any such ϵ to a suitable $N(\epsilon)$.
- In the case of $a_n = \frac{1}{n}$ for all $n \geq 1$, this is straightforward.

For all $\epsilon > 0$



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

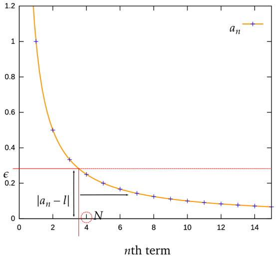
- We apply the limit inequality in general and derive a condition for n.

$$\frac{1}{n} < \epsilon \Leftrightarrow n > \frac{1}{\epsilon}.$$

- We are looking for a greater-than inequality. So long as we get one, we can select the next largest integer value; we use the ceiling function to do this, giving

$$N(\epsilon) = \left\lceil \frac{1}{\epsilon} \right\rceil.$$ 取里里面的值比他大，离他最近的整数

We are done!



The convergence of the sequence $a_n = 1/n$ for $n \geq 1$ tending to 0.

- Now whatever value of ϵ we are given, we can find a value of N using this function and we are done.
- The notation $N(\epsilon)$ in

$$N(\epsilon) = \left\lceil \frac{1}{\epsilon} \right\rceil$$

reminds us that the value of N is indeed a function of the chosen value of ϵ.

Proving divergence

- How do you prove that $((-1)^n)_{n \geq 1}$ diverges?

Proving divergence (i)

- First, we'll prove that this sequence does not converge to a limit l.
- We will produce a proof by contradiction.
- Assume (for a contradiction) that $((-1)^n)_{n\geq 1}$ converges to l.
- Then for every $\epsilon > 0$, we can find a N in $\mathbb{N}$ such that, for all $n > N$, $|a_n - l| < \epsilon$.
- Since this must hold for every ϵ, we can pick a particular ϵ: let's pick $\epsilon = 1$.

Proving divergence (ii)

- Pick some $n > N$. We have
$$|a_{2n} - l| < 1.$$
- But
$$|a_{2n} - l| = |(-1)^{2n} - l| = |1 - l| < 1.$$
- This can be written as
$$-1 < 1 - l < 1.$$
- Hence
$$0 < l < 2.$$

Proving divergence (iii)

- We also have
$$|a_{2n+1} - l| < 1.$$
- But
$$|a_{2n+1} - l| = |(-1)^{2n+1} - l| = |-1 - l| < 1.$$
- This can be written as
$$-1 < -1 - l < 1.$$
- Hence
$$-2 < l < 0.$$

Proving divergence (iv)

- But the inequalities $0 < l < 2$ and $-2 < l < 0$ contradict each other—there is no such l that satisfies them both.
- We have arrived at a contradiction—therefore our assumption that $((-1)^n)_{n\geq 1}$ converges to a limit l is false.
- The opposite must be true: $((-1)^n)_{n\geq 1}$ does not converge to a limit.

Are we there yet?

- So have we proved that $((-1)^n)_{n\geq 1}$ diverges?

Not yet...

- We still need to show that $((-1)^n)_{n\geq 1}$ does not converge to ∞ and does not converge to $-\infty$.
- How would you go about it?

Look at the definitions

- Let's look at the definitions:
  - The sequence $(a_n)_{n\geq 1}$ converges to ∞ (in the extended real line $\mathbb{R} \cup \{\infty\}$) iff for all r in $\mathbb{R}$, there is an N in $\mathbb{N}$ such that for all $n \geq N$ we have $a_n > r$. We write $\lim_{n\to\infty} a_n = \infty$ or $(a_n)_{n\geq 1} \to \infty$ in that case.
  - The sequence $(a_n)_{n\geq 1}$ converges to $-\infty$ (in the extended real line $\mathbb{R} \cup \{-\infty\}$) iff for all r in $\mathbb{R}$, there is an N in $\mathbb{N}$ such that for all $n \geq N$ we have $a_n < r$. We write $\lim_{n\to\infty} a_n = -\infty$ or $(a_n)_{n\geq 1} \to -\infty$ in that case.
- Pick $r = 2$ in the first definition. We know that no matter what n, $a_n \leq 2$. Therefore $(a_n)_{n\geq 1}$ does not converge to ∞.
- Pick $r = -2$ in the second definition. We know that no matter what n, $a_n \geq -2$. Therefore $(a_n)_{n\geq 1}$ does not converge to $-\infty$.
- Since $((-1)^n)_{n\geq 1}$ does not converge to a limit l, does not converge to ∞, and does not converge to $-\infty$—it does not converge. Therefore, it diverges.
- We are done: Q.E.D. (Latin for *quod erat demonstrandum*—"which was to be demonstrated").
- Since Q.E.D. is a bit pretentious, we'll use the Halmos symbol instead: $\square$

Common convergent sequences

1. $(a_n)_{n\geq 1} = \left(\frac{1}{n}\right)_{n\geq 1} \to 0$, also $(a_n)_{n\geq 1} = \left(\frac{1}{n^2}\right)_{n\geq 1} \to 0$, $(a_n)_{n\geq 1} = \left(\frac{1}{\sqrt{n}}\right)_{n\geq 1} \to 0$
2. In general, $(a_n)_{n\geq 1} = \left(\frac{1}{n^c}\right)_{n\geq 1} \to 0$ whenever c is a positive real constant $c > 0$
3. $(a_n)_{n\geq 1} = \left(\frac{1}{2^n}\right)_{n\geq 1} \to 0$, also $(a_n)_{n\geq 1} = \left(\frac{1}{3^n}\right)_{n\geq 1} \to 0$, $(a_n)_{n\geq 1} = \left(\frac{1}{e^n}\right)_{n\geq 1} \to 0$
4. In general, $(a_n)_{n\geq 1} = \left(\frac{1}{c^n}\right)_{n\geq 1} \to 0$ whenever c is a real constant with $|c| > 1$;
   or equivalently $(a_n)_{n\geq 1} = (c^n)_{n\geq 1} \to 0$ whenever c is a real constant with $|c| < 1$
5. $(a_n)_{n\geq 1} = \left(\frac{1}{n!}\right)_{n\geq 1} \to 0$
6. $(a_n)_{n\geq 1} = \left(\frac{1}{\log n}\right)_{n\geq 1} \to 0$

Combinations of sequences

- It is fortunate that the convergence of sequences to limits enjoys good compositionality properties.
- By this we mean that if sequence elements get combined by some algebraic operator such as addition, and the argument sequences converge to a limit, then the combined sequence will converge to the combined limits.
- Given sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ which converge to limits a and b respectively, and a real constant λ, we have
  1. $\lim_{n\to\infty} \lambda a_n = \lambda a$
  2. $\lim_{n\to\infty} a_n + b_n = a + b$
  3. $\lim_{n\to\infty} a_n - b_n = a - b$
  4. $\lim_{n\to\infty} a_n b_n = ab$
  5. $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$ provided that $b \neq 0$
- Note that the first item refers to a sequence $(\lambda a_n)_{n\geq 1}$ and that its limit equals λ times the limit of sequence $(a_n)_{n\geq 1}$.
- The other items have similar interpretations.

Triangle inequality

- Let us prove one of these results, namely $\lim_{n\to\infty} a_n + b_n = a + b$.
- We will need a lemma—the triangle inequality: For any real numbers a and b

$$|a + b| \leq |a| + |b|.$$

- In order to prove the lemma, we proceed as follows:

$$\begin{aligned} |a+b|^2 = (a+b)^2 &= a^2 + 2ab + b^2 \\ &= |a|^2 + 2ab + |b|^2 \\ &\leq |a|^2 + 2|ab| + |b|^2 \\ &= |a|^2 + 2|a||b| + |b|^2 \\ &= (|a| + |b|)^2. \end{aligned}$$

- We can show (do it!) that if x and y are positive, then $x \leq y$ iff $x^2 \leq y^2$.
- Therefore

$$|a + b| \leq |a| + |b|.$$

- We are done: $\square$

How to prove it?

- Now for the main result: Given sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ which converge to limits a and b respectively, and a real constant λ, we have $\lim_{n\to\infty} a_n + b_n = a + b$.
- Let $\epsilon > 0$ be given. Then $\frac{1}{2}\epsilon > 0$. Since $a_n \to a$ as $n \to \infty$, it follows that we can find an N_1 such that, for any $n > N_1$,
$$|a_n - a| < \frac{1}{2}\epsilon. \tag{2}$$
- Similarly, we can find an N_2 such that, for any $n > N_2$,
$$|b_n - b| < \frac{1}{2}\epsilon. \tag{3}$$
- Let N be whichever of N_1 and N_2 is the larger, i.e. $N = \max\{N_1, N_2\}$. Then, if $n > N$, both inequalities (2) and (3) are true simultaneously. Thus, for any $n > N$,
$$\begin{aligned} |(a_n + b_n) - (a + b)| &= |(a_n - a) + (b_n - b)| \\ &\leq |a_n - a| + |b_n - b| \quad \text{(triangle inequality)} \\ &< \frac{1}{2}\epsilon + \frac{1}{2}\epsilon = \epsilon. \end{aligned}$$
- Given any $\epsilon > 0$ we have found a value of N (namely $N = \max\{N_1, N_2\}$) such that, for any $n > N$, $|(a_n + b_n) - (a + b)| < \epsilon$. Hence $a_n + b_n \to a + b$ as $n \to \infty$. $\square$

Example (i)

- These sequence combination results are useful for being able to understand convergence properties of combined sequences.
- As an example, if we take the sequence defined by

$$a_n = \frac{3n^2 + 2n}{6n^2 + 7n + 1}$$

for all $n \geq 1$, then we can transform this sequence into a combination of sequences that we know how to deal with, by dividing the numerator and denominator by n^2:

$$a_n = \frac{3 + \frac{2}{n}}{6 + \frac{7}{n} + \frac{1}{n^2}} = \frac{b_n}{c_n},$$

where

$$b_n = 3 + \frac{2}{n} \text{ and } c_n = 6 + \frac{7}{n} + \frac{1}{n^2}.$$

- We know from rule (5) above that if $(b_n)_{n\geq 1} \to b$ and $(c_n)_{n\geq 1} \to c \neq 0$, then

$$\left(\frac{b_n}{c_n}\right)_{n\geq 1} \to \frac{b}{c}$$

- This means that we can investigate the convergence of $(b_n)_{n\geq 1}$ and $(c_n)_{n\geq 1}$ separately.

Example (ii)

- Similarly we can rewrite b_n as $3 + d_n$, for $d_n = \frac{2}{n}$.
- We know that $(d_n)_{n\geq 1} = (\frac{2}{n})_{n\geq 1} \to 0$ as it is one of the common sequence convergence results, thus $(b_n)_{n\geq 1} \to 3$ by rule (2) above.
- By a similar argument, we can see that $(c_n)_{n\geq 1} \to 6$ using composition rule (2) and common results.
- Finally, we get the result that
$$(a_n)_{n\geq 1} \to \frac{3}{6} = \frac{1}{2}.$$

Some issues

- The definition of a sequence $(a_n)_{n\geq 1}$ to have a limit l is certainly useful, as we have already appreciated above.
- However, there are two issues with this that we now want to resolve:
  - Uniqueness of limit: What if a sequence could converge to more than one limit? In other words, is it possible that a sequence converges to more than one limit?
  - Convergence as guessing: Our definition of convergence assumes that we guessed the value of a limit, which may be a hard undertaking. Can we come up with a definition of convergence that appeals only to the sequence itself and does not require such guesswork?

Uniqueness of limits

- How would you show that a sequence $(a_n)_{n \geq 1}$ can only have one limit?

Proof by contradiction

- Let $(a_n)_{n\geq 1}$ denote a sequence with more than one limit, two of which are labelled as l_1 and l_2 with $l_1 \neq l_2$.
- Choose $\epsilon = \frac{1}{3}|l_1 - l_2|$ which is greater than zero since $l_1 \neq l_2$.
- Since l_1 is a limit of $(a_n)_{n\geq 1}$, we can apply the definition of limit with our choice of ϵ to find $N_1 \in \mathbb{N}$ such that
$$|a_n - l_1| < \epsilon \text{ for all } n \geq N_1.$$
- Similarly, as l_2 is a limit of $(a_n)_{n\geq 1}$, we can apply the definition of limit with our choice of ϵ to find $N_2 \in \mathbb{N}$ such that
$$|a_n - l_2| < \epsilon \text{ for all } n \geq N_2.$$
- Choose any $m_0 > \max\{N_1, N_2\}$, then $|a_{m_0} - l_1| < \epsilon$ and $|a_{m_0} - l_2| < \epsilon$.
- This shows that l_1 is "close to" a_{m_0} and l_2 is also "close to" a_{m_0}. Hence we must have that l_1 is "close to" l_2:
$$\begin{aligned} |l_1 - l_2| &= |l_1 - a_{m_0} + a_{m_0} - l_2| \quad \text{(trick: "adding in zero")} \\ &\leq |l_1 - a_{m_0}| + |a_{m_0} - l_2| \quad \text{(triangle inequality)} \\ &< \epsilon + \epsilon \\ &= 2\epsilon = \frac{2}{3}|l_1 - l_2|. \end{aligned}$$
- So we find that $|l_1 - l_2| < \frac{2}{3}|l_1 - l_2|$, which is a contradiction. $\square$

Eliminating guessing

- If we want to get rid of having to guess the limit for a definition of convergence, we will require a somewhat more complex definition, since we no longer can work with a claimed limit value $l \in \mathbb{R}$.
- The notion of a Cauchy sequence turns out to be the appropriate concept.
- Its intuition is that we keep the roles of ϵ and n as in the definition with a limit l.
- But that then we need to argue that all points in the sequence that come after a_N are within ϵ of each other.

Cauchy sequences

- A sequence $(a_n)_{n \geq 1}$ is a Cauchy sequence iff for all $\epsilon > 0$, there is some N in $\mathbb{N}$ such that for all $n, m > N$ we have $|a_n - a_m| < \epsilon$.
- It is not sufficient for each term to become arbitrarily close to the preceding term. For instance, in the sequence of square roots of natural numbers

  $$a_n = \sqrt{n}$$

  the consecutive terms become arbitrarily close to each other:

  $$a_{n+1} - a_n = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}.$$

  However, with growing values of the index n, the terms a_n become arbitrarily large. So for any index n and distance d there exists an index m big enough such that $a_m - a_n > d$. (Actually, any $m > (\sqrt{n} + d)^2$ suffices.) As a result, despite how far one goes, the remaining terms of the sequence never get close to each other, hence the sequence is not Cauchy.

Question

- Can you show that every sequence that converges to a real number is a Cauchy sequence?

Every sequence that converges to a real number is a Cauchy sequence

- Let $(a_n)_{n\geq 1} \to l$ as $n \to \infty$.
- Given $\epsilon > 0$, there exists N_1 such that, for all $n \geq N_1$, $|a_n - l| < \frac{\epsilon}{2} < \epsilon$.
- For $m > n \geq N_1$ we also have $|a_m - l| < \frac{\epsilon}{2} < \epsilon$.
- Let $N \geq N_1$, then for all $n, m \geq N$ we have

$$\begin{aligned}
|a_n - a_m| &= |a_n - l - a_m + l| \\
&\leq |a_n - l| + |-(a_m - l)| \quad \text{(triangle inequality)} \\
&= |a_n - l| + |a_m - l| \\
&< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

- Therefore $(a_n)_{n\geq 1}$ is a Cauchy sequence.

An interesting example

- Here is an example of a sequence that converges to a real number and is thus Cauchy.
- Let $a_1 = 1$ and set $a_{n+1} = \frac{1}{2} \cdot (a_n + 2/a_n)$ for all $n \geq 1$.
- This defines a sequence of rational numbers that converges to $\sqrt{2}$ (the proof is rather difficult).
- In particular, it follows from the uniqueness of limits that the Cauchy sequence in the above example does not have a limit over the rational numbers $\mathbb{Q}$.

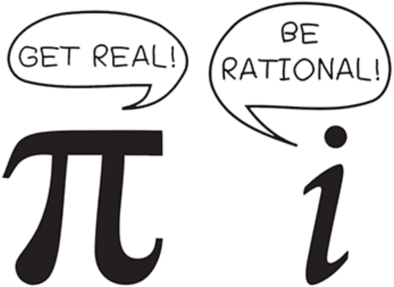
Completeness

- A subset $A \subseteq \mathbb{R}$ is said to be complete if any Cauchy sequence in A converges to a limit in A.

Rationals are not complete; reals are

- This means that the rational numbers are not complete since not all Cauchy sequences of rational numbers have limits in $\mathbb{Q}$.
- In contrast, the set $\mathbb{R}$ of real numbers is complete as we will prove later.

Differences between number systems...



Cauchy sequences in $\mathbb{R}^n$ and other metric spaces

- Cauchy sequences can be defined similarly when elements a_n are vectors and the expression $|a_n - a_m|$ refers to a metric that generalizes the absolute value.
- The real vector spaces that we cover in Linear Algebra are then also complete in this sense.

An overview of the sandwich theorem

- The sandwich theorem is an alternative, and often a simpler, way of proving that a sequence converges to a limit.
- In order to use the sandwich theorem, you need to have two sequences (one of which can be constant) which bound the sequence you wish to reason about and which converge to the same limit.
- The two bounding sequences should be given as known results, or proved to converge prior to the use of the sandwich theorem.
- The bounding sequences should be easier to reason about than the sequence that we want to reason about.

The sandwich theorem

Let $(l_n)_{n\geq 1}$ and $(u_n)_{n\geq 1}$ be sequences, and l a real number where both $\lim_{n\to\infty} l_n = l$ and $\lim_{n\to\infty} u_n = l$. Suppose that for a third sequence $(a_n)_{n\geq 1}$ we have:

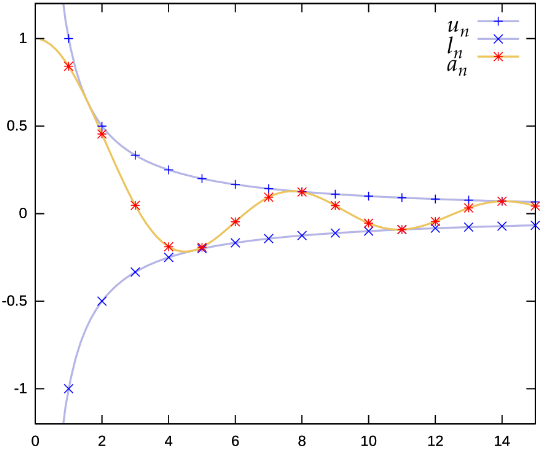
$$\text{there is some } N \in \mathbb{N} \text{ such that } l_n \leq a_n \leq u_n \text{ for all } n \geq N$$

Then $\lim_{n\to\infty} a_n = l$ as well.

Example: applying the sandwich theorem

- Consider the sequence
$$(a_n)_{n\geq 1} = \left(\frac{\sin n}{n}\right)_{n\geq 1},$$
which oscillates around its limit of 0.
- We construct two sandwiching sequences where $l_n = -1/n$ and $u_n = 1/n$.
- Since we know (or can easily prove) that $(l_n)_{n\geq 1}$ and $(u_n)_{n\geq 1}$ both tend to 0, we only need to show that there is some N in $\mathbb{N}$ such that $l_n \leq a_n$ and $a_n \leq u_n$ from all n with $n > N$.
- In our particular case, we can show this for all $n > 0$: we can prove this in one line by referring to a property of the sin function:
$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n} \Leftrightarrow -1 \leq \sin n \leq 1,$$
the inequalities on the right-hand side being true for all n by the range of $\sin n$.

Demonstration of the sandwich theorem



The sequence $(a_n)_{n\geq 1}$ is sandwiched between two bounding sequences $(l_n)_{n\geq 1}$ and $(u_n)_{n\geq 1}$ which converge to the same limit 0.

Proof of the the sandwich theorem (i)

- We know that $(l_n)_{n\geq 1}$ and $(u_n)_{n\geq 1}$ tend to the same limit l.
- We use the definition of convergence for both of these sequences.
- Given some $\epsilon > 0$, we can find an N_1 and N_2 such that for all $n > N_1$, $|l_n - l| < \epsilon$ and for all $n > N_2$, $|u_n - l| < \epsilon$.
- If we want both sequences to be simultaneously less than ϵ from the limit l for a single value of N', then we have to pick $N' = max(N_1, N_2)$.
- So let n be such that $n > N'$. Then we know that

$$|l_n - l| < \epsilon \text{ and } |u_n - l| < \epsilon.$$

- Therefore, this together with the fact that $l_n < u_n$ gives us:

$$l - \epsilon < l_n < u_n < l + \epsilon.$$

Proof of the the sandwich theorem (ii)

- By the assumption of the sandwich theorem, there is some N in $\mathbb{N}$ such that $l_n \leq a_n \leq u_n$ for all $n > N$. Hence, for all $n > \max(N, N')$ we conclude that:

$$\begin{aligned} l - \epsilon < l_n \leq a_n \leq u_n < l + \epsilon \Leftrightarrow \\ l - \epsilon < a_n < l + \epsilon \Leftrightarrow \\ -\epsilon < a_n - l < \epsilon \Leftrightarrow \\ |a_n - l| < \epsilon. \end{aligned}$$

- Since n was an arbitrary natural number larger than $\max(N, N')$, this proves that $(a_n)_{n\geq 1}$ converges to l as a result.
- We note that $N(\epsilon)$ we determined here is $\max(N, N')$, which combines two such "N" values from the convergence of the two bounding sequences.

Ratio tests for sequences

- A third technique (and a very simple one) for proving sequence convergence is called the ratio test.
- Its name stems from the fact that it compares the ratio of consecutive terms in the sequence.
- The test can be used to verify whether a sequence converges to 0 or diverges.
- This is not really restricting its usage: If a sequence $(b_n)_{n\geq 1}$ has a nonzero limit, $l \neq 0$, then the ratio test can still be applied to a modified sequence $(a_n)_{n\geq 1}$ where a_n equals $b_n - l$ for all $n \geq 1$.
- If the modified sequence passes the test, we know it converges to 0 and so sequence $(b_n)_{n\geq 1}$ converges to l.

The ratio test for sequences

Let c in $\mathbb{R}$ be such that $0 \leq c < 1$. Suppose that there is some N in $\mathbb{N}$ such that for all $n \geq N$ we have $\left|\frac{a_{n+1}}{a_n}\right| \leq c$. Then $\lim_{n\to\infty} a_n = 0$.

Example

- Consider the sequence $(a_n)_{n\geq 1}$ where a_n equals 3^{-n} for all $n \geq 1$.
- We have

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{3^{-(n+1)}}{3^{-n}}\right| = \frac{1}{3} < 1.$$

- So we can choose c to be $\frac{1}{3}$ and N to be 1 and the ratio test for sequences passes for these values.
- Therefore, $(3^{-n})_{n\geq 1}$ has limit 0.

An overview of the limit ratio test

- The limit ratio test is a method that is derived from the ratio test that we just discussed.
- It works by considering the limit of the ratio of consecutive terms in the sequence.
- This assumes that this limit exists.
- This is often the case when the above ratio test is inconclusive, e.g., when the ratios have n as a variable after any simplifications and so guessing a suitable constant c may be hard.

The limit ratio test

Suppose that the limit

$$r = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists. If $r < 1$, then the sequence $(a_n)_{n \geq 1}$ converges to 0.

Example: ratio test and limit ratio test

- Consider the sequence $(a_n)_{n\geq 1}$ where a_n equals $\frac{1}{n!}$ for all $n \geq 1$. We compute

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{1/(n+1)!}{1/n!}\right| = \frac{1}{n+1}.$$

- Since $c = \frac{1}{2}$ satisfies $0 \leq c < 1$, and since for all $n > 1$, we have that $1/(n+1) \leq c$, we may apply the ratio test for $c = \frac{1}{2}$ and $N = 1$ to conclude that the sequence $(1/n!)_{n\geq 1}$ converges to 0.
- We can also apply the limit ratio test here since

$$\lim_{n\to\infty} \left(\frac{1}{n+1}\right)_{n\geq 1} = 0.$$

- So $r < 1$ since $r = 0$. Therefore, the limit ratio test passes and we also conclude with this test that this sequence has limit 0.

Proof of the correctness of the ratio test (i)

- We are given c in $\mathbb{R}$ with $c < 1$ and N in $\mathbb{N}$ such that $\left|\frac{a_{n+1}}{a_n}\right| \leq c < 1$ for all $n \geq N$.
- For the left-hand inequality, we multiply both sides by $|a_n|$ to get $|a_{n+1}| \leq c|a_n|$ for $n \geq N$.
- Similarly, we get $|a_n| \leq c|a_{n-1}|$ and so forth.
- In fact, repeating this creates a chain of inequalities down to the point when n equals N, at which point that inequality also holds. Therefore, we obtain:

$$|a_n| \leq c|a_{n-1}| \leq c^2|a_{n-2}| \leq \ldots \leq c^{n-N}|a_N|$$

  for all $n \geq N$.
- In particular, this means that for all $n \geq N$ we have

$$|a_n| \leq c^{n-N}|a_N| \leq kc^n \text{ where } k = \frac{|a_N|}{c^N} \text{ is a constant.}$$

Proof of the correctness of the ratio test (ii)

- Now, since $c < 1$, we know that the sequence $(b_n)_{n\geq 1}$ with $b_n = kc^n$ for all $n \geq 1$ has limit 0, by the standard result (4) from our common convergent sequences.
- We have bounded the sequence $(a_n)_{n\geq 1}$ between $(-b_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$.
- Since both of these sequences have limit 0, we can apply the sandwich theorem to conclude that the sequence $(a_n)_{n\geq 1}$ converges to 0. □

Proof of the correctness of the limit ratio test (i)

- We are given that the new sequence $(b_n)_{n\geq 1}$ with $b_n = \left|\frac{a_{n+1}}{a_n}\right|$ has limit r. Recall that this means that for all $\epsilon > 0$, we can find an N in $\mathbb{N}$ such that for all $n > N$ we have $|b_n - r| < \epsilon$.
- When we unpack that last inequality, we get:

$$r - \epsilon < b_n < r + \epsilon. \tag{4}$$

- We will now pick specific values of ϵ and then reason about the inequalities (4) in order to reduce the limit ratio test to the corresponding test without the limit.
- Assume $r < 1$. Recall that we have to prove that the sequence $(a_n)_{n\geq 1}$ converges.
- Since $r < 1$, we know that ϵ defined as $\frac{1-r}{2}$ is positive.
- Therefore, there is some $N(\epsilon)$ such that for all $n \geq N$ the inequalities in (4) are true, and they now read as

$$r - \frac{1-r}{2} < b_n < r + \frac{1-r}{2}$$

  which we can simplify to

$$\frac{3r-1}{2} < b_n < \frac{r+1}{2}.$$

Proof of the correctness of the limit ratio test (ii)

- Taking the right-hand side, we see that $b_n < \frac{r+1}{2}$, and, since $r < 1$, we can show that $\frac{r+1}{2} < 1$ also.
- If we take $c = \frac{r+1}{2} < 1$ and N to be the $N(\epsilon)$ above, then the original ratio test passes for these choices.
- Since we have proved the correctness of the ratio test already, we can conclude that the original sequence $(a_n)_{n \geq 1}$ converges, as claimed. $\square$

Subsequences

- If $f : \mathbb{N} \to \mathbb{R}$ is a sequence and $M \subseteq \mathbb{N}$ an infinite subset, then the restriction $f : M \to \mathbb{R}$ is called a subsequence of f.
- Using the usual notation $(a_n)_{n\geq 1}$ for a sequence, any subsequence of this sequence would be of the form $(a_{n_i})_{i\geq 1}$ where n_i are positive integers with $n_1 < n_2 < n_3 < \ldots$, i.e., $M = \{n_i : i \geq 1\}$.

Examples

- Suppose $a_n = 1/n$.
- Then $a_{n_i} = 1/(2i)$ is the subsequence $1/2, 1/4, 1/6, \ldots$ of even terms.
- And $a_{n_i} = 1/(3i-2)$ is the subsequence $a_1, a_4, a_7, a_{10}, \ldots$.

Theorem

- Any subsequence of a convergence sequence converges to the limit of the sequence.

Proof

- Suppose $\lim_{n\to\infty} a_n = l$, where l is a real number.
- Let $(a_{n_i})_{i\geq 1}$ be any subsequence.
- Let $\epsilon > 0$ be given.
- Since $\lim_{n\to\infty} a_n = l$, there exists $N > 0$ such that for $n > N$ we have $|a_n - l| < \epsilon$.
- Note that by definition we always have $n_i \geq i$.
- Therefore for $i > N$, we have $|a_{n_i} - l| < \epsilon$.
- The proof for sequences that converge to $\pm\infty$ is similar. $\square$

Theorem

- Any sequence of real numbers has a monotonic subsequence.

Proof

- Consider a sequence $(a_n)_{n\geq 1}$.
- For any $m \geq 1$, we say that a_m is a peak of $(a_n)_{n\geq 1}$ if $a_m \geq a_n$ for all $n \geq m$.
- (A strictly increasing sequence will have no peaks while in a decreasing sequence every term is a peak.)
- There are now two cases.
  - Suppose $(a_n)_{n\geq 1}$ has infinitely many peaks denoted by $a_{n_1}, a_{n_2}, \ldots$. Then the subsequence $(a_{n_i})_{i\geq 1}$ is monotonically decreasing.
  - If, on the other hand, $(a_n)_{n\geq 1}$ has only a finite number of peaks $a_{n_1}, a_{n_2}, \ldots, a_{n_k}$, then choose $t_1 = n_k + 1$. Since t_1 is not a peak, there exists $n > t_1$ such that $t_2 := n$ satisfies $a_{t_1} < a_{t_2}$. In this way construct t_i for all $i \geq 1$ such that $(a_{t_i})_{i\geq 1}$ is a monotonically increasing sequence. $\square$

Useful techniques

- A commonly used decomposition of the absolute value operator is

$$|x| < a \Leftrightarrow -a < x < a.$$

- Where we are combining absolute values of expressions, the following are useful rules, where x and y are real numbers:

Abs1 $|x \cdot y| = |x| \cdot |y|$.
Abs2 $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$.
Abs3 $|x + y| \leq |x| + |y|$, the triangle inequality.
Abs4 $|x - y| \geq ||x| - |y||$, the reverse triangle inequality.

The reverse triangle inequality

- We have already seen a proof of the triangle inequality.
- Let us prove the reverse triangle inequality.
- From the triangle inequality it follows that

$$|y| = |x + y - x| \leq |x| + |y - x|$$

  and

$$|x| = |y + x - y| \leq |y| + |x - y|.$$

- Hence

$$|y - x| \geq |y| - |x| \text{ and } |x - y| \geq |x| - |y|.$$

- From Abs1, $|y - x| = |(-1)(x - y)| = |-1||x - y| = |x - y|$. So we have

$$|x - y| \geq -(|x| - |y|) \text{ and } |x - y| \geq |x| - |y|.$$

- Hence

$$|x - y| \geq ||x| - |y||,$$

  as required. □

Properties of real numbers

- Now that we have looked at limits of sequences of real numbers, we will see how this relates to some fundamental properties of sets of real numbers.
- Such considerations will lead us to describe the so-called fundamental axiom of analysis.
- Both the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ have a linear ordering $<$, meaning that for numbers x and y we either have $x < y$, $y < x$, or $x = y$.
- This allows us to reveal an intimate connection between this order structure and the convergence behaviour of certain sequences.
- For that, we need to review concepts from order theory. We do this for real numbers, the definitions for rational numbers are essentially the same ones.

Concepts from order theory

Let X be a set of real numbers.

1. Let l and u be real numbers. Then
   1.1 u is an upper bound of X if $x \leq u$ for all $x \in X$;
   1.2 l is a lower bound of X if $l \leq x$ for all $x \in X$;
   1.3 a least upper bound (supremum, $\sup(X)$) of X is an upper bound s of X such that $s \leq u$ for all upper bounds u of X;
   1.4 a greatest lower bound (infimum, $\inf(X)$) of X is a lower bound i of X such that $l \leq i$ for all lower bounds l of X.
2. We say that such a set X is
   2.1 bounded above if X has an upper bound;
   2.2 bounded below if X has a lower bound;
   2.3 bounded if X has an upper and lower bound.

Uniqueness of the least upper bound

- The least upper bound of a set X is unique if it exists.
- How would you prove that?

Proof

- Let $L, L' \in \mathbb{R}$ such that L, L' are both least upper bounds of X.
- Because L is an upper bound of X and L' is a least upper bound of X,

$$L' \leq L.$$

- Because L' is an upper bound of X and L' is a least upper bound of X,

$$L \leq L'.$$

- Thus $L = L'$.
- The uniqueness of the greatest lower bound (if it exists) is proved similarly.

Example

- The set of positive real numbers $\{x \in \mathbb{R} \mid x > 0\}$ does not have any upper bounds and so also no least upper bound.
- But it has 0 and all negative numbers as lower bounds and 0 is its greatest lower bound.

Dedekind-completeness of $\mathbb{R}$

- Axiom of Dedekind-completeness for real numbers: Every nonempty subset X of the real numbers $\mathbb{R}$ that is bounded above has a least upper bound.

From this we can derive...

- We are now in a position to state that a monotonically increasing sequence that is bounded above has a limit.

Fundamental theorem of analysis

Let $(a_n)_{n\geq 1}$ be a sequence of real numbers that is

1. increasing, i.e., for all $n > m \geq 1$ we have $a_n \geq a_m$, and
2. bounded above, i.e., there is some u in $\mathbb{R}$ such that $a_n \leq u$ for all $n \geq 1$.

Then $s = \sup\{a_n \mid n \geq 1\}$ exists and is the limit of the sequence $(a_n)_{n\geq 1}$.

Proof

- By the axiom of Dedekind-completeness, we know that the supremum s stated in the theorem exists since $\{a_n \mid n \geq 1\}$ is nonempty and bounded above.
- We need to show that s is the limit of that sequence.
- So let $\epsilon > 0$ be given.
- We prove this in stages...

CLAIM 1: There exists some N in $\mathbb{N}$ such that $|a_N - s| < \epsilon$

- Proof by contradiction.
- If this claim is false, then we have $|a_n - s| \geq \epsilon$ for all $n \geq 1$.
- This implies $s - a_n \geq \epsilon$, i.e. $s - \epsilon \geq a_n$ for all $n \geq 1$ since $s \geq a_n$.
- But then $s - \epsilon$ is an upper bound of the set $\{a_n \mid n \geq 1\}$.
- Since s is the least such upper bound, we get that $s \leq s - \epsilon$.
- But this is a contradiction as ϵ is positive.

CLAIM 2: For the N whose existence we proved in CLAIM 1, we have that $|a_n - s| < \epsilon$ for all $n > N$

- Since the sequence is monotonically increasing, we have that $n > N$ implies $a_n \geq a_N$.
- And so

  $$0 \leq s - a_n \leq s - a_N < \epsilon$$

  proves the claim, where the first inequality follows since s is an upper bound of $\{a_n \mid n \geq 1\}$ and the second inequality follows from CLAIM 1.
- By the definition of convergence, this shows that $(a_n)_{n \geq 1}$ has limit s.

Similarly...

...a bounded below decreasing sequence is convergent to a real number.

Do you remember...

...the definition of completeness?

Completeness

- A subset $A \subseteq \mathbb{R}$ is said to be complete if any Cauchy sequence in A converges to a limit in A.

The set of real numbers is complete

- Let $(a_n)_{n\geq 1}$ be a Cauchy sequence.
- We show that it converges to a limit in $\mathbb{R}$ from which the result follows.

CLAIM 1: $(a_n)_{n\geq 1}$ is bounded

- Since the sequence is Cauchy, putting $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that for $n, m > N$ we have $|a_n - a_m| < 1$.
- Put $K = 1 + \max\{|a_1|, |a_2|, \ldots, |a_N|, |a_{N+1}|\}$.
- Then for $m > N$ we have $|a_m| \leq |a_{N+1}| + |a_m - a_{N+1}| < |a_{N+1}| + 1$.
- Hence, $|a_n| < K$ for all $n \geq 1$.

CLAIM 2: The sequence $(a_n)_{n\geq 1}$ has a convergent subsequence

- Since any sequence of real numbers has a monotonic subsequence, $(a_n)_{n\geq 1}$ has a monotonic subsequence $(a_{n_i})_{i\geq 1}$, say, which is also bounded (by CLAIM 1).
- Hence, by the fundamental theorem of analysis, the subsequence is convergent with, say, $\lim_{i\to\infty} a_{n_i} = l$.

CLAIM 3: $\lim_{n\to\infty} a_n = l$

- Let $\epsilon > 0$ be given.
- Since $(a_n)_{n\geq 1}$ is Cauchy, there exists $N_1 \in \mathbb{N}$ such that for $n, m > N_1$ we have $|a_n - a_m| < \epsilon/2$.
- Since $\lim_{i\to\infty} a_{n_i} = l$, there exists $N_2 > 0$ such that for $i > N_2$, we have $|a_{n_i} - l| < \epsilon/2$.
- Let $N = \max\{N_1, N_2\}$.
- Then

  $$|a_n - l| = |a_n - a_{n_i} + a_{n_i} - l| \leq |a_n - a_{n_i}| + |a_{n_i} - l| \leq \epsilon/2 + \epsilon/2 = \epsilon$$

  for any $n, i > N$.
- Thus, $|a_n - l| < \epsilon$ for $n > N$.
- Hence, $(a_n)_{n\geq 1}$ is convergent to a real number.

Question

- We already know that the set $\mathbb{Q}$ of rationals is not (Cauchy-) complete.
- Can you find some other subsets of $\mathbb{R}$ that are complete?

An answer

- For example, closed intervals of the form $[a, b]$, $a < b$.
- What about $\mathbb{Z}$ and $\mathbb{N}$?

The integers and natural numbers are Cauchy-complete

- Let $(a_n)_{n\geq 1}$ be a Cauchy sequence of integers.
- Let $\epsilon = 1$.
- There exists $N \in \mathbb{N}$ such that for all $n, m > N$ we have $|a_n - a_m| < \epsilon = 1$.
- But this is only possible (for integers) if $a_n = a_m$.
- Thus $(a_n)_{n\geq 1}$ is eventually constant, i.e., for all $n > N$ we have $a_n = a$ for some $a \in \mathbb{Z}$.
- Therefore, $\lim_{n\to\infty} a_n = a \in \mathbb{Z}$.
- Hence the integers are Cauchy-complete.
- The same proof applies to the natural numbers, which are thus also Cauchy-complete.